

SQL Interview Questions – Day 1

5 beginner-friendly interview questions with concise answers

1. What is SQL?

Answer: SQL (Structured Query Language) is a language used to store, retrieve, manipulate, and manage data in relational databases such as MySQL, PostgreSQL, and SQL Server.

2. What is the difference between WHERE and HAVING?

Answer: WHERE filters individual rows before grouping, while HAVING filters groups after GROUP BY is applied.

```
SELECT department, COUNT(*) AS total
FROM employees
WHERE salary > 30000
GROUP BY department
HAVING COUNT(*) > 5;
```

3. What is a Primary Key?

Answer: A primary key is a column, or combination of columns, that uniquely identifies each row in a table. It cannot contain NULL values, and each value must be unique.

```
CREATE TABLE students (
    student_id INT PRIMARY KEY,
    name VARCHAR(50)
);
```

4. What is the difference between INNER JOIN and LEFT JOIN?

Answer: INNER JOIN returns only rows that have matching values in both tables. LEFT JOIN returns all rows from the left table and the matching rows from the right table; unmatched right-side values become NULL.

```
SELECT e.name, d.department_name
FROM employees e
LEFT JOIN departments d
    ON e.department_id = d.department_id;
```

5. What is GROUP BY used for?

Answer: GROUP BY groups rows with the same values so aggregate functions such as COUNT(), SUM(), AVG(), MAX(), and MIN() can be applied to each group.

```
SELECT department, AVG(salary) AS avg_salary
FROM employees
GROUP BY department;
```